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GENERATING DYNAMIC MULTI-DISTRIBUTION INVERSION MODELS TO FACILITATE WELLBORE OPERATIONS

Abstract

A system can provide a dynamic multi-distribution model to facilitate a wellbore operation. For example, the system can receive, from a downhole tool deployed in a wellbore during a drilling operation, resistivity data for a geological formation associated with an interval of the wellbore. The system can further execute a resistivity inversion algorithm to generate distribution outputs of a resistivity inversion model using the resistivity data. Additionally, the system can select a section of each of the distribution outputs. Each section of each distribution output can correspond with a different segment of the interval of the wellbore. The system can then generate a dynamic multi-distribution model. The dynamic multi-distribution model can include the sections selected from each of the distribution outputs. The system can further output, by a user interface, the dynamic distribution output, which can be used to adjust the drilling operation.

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Background/Summary

TECHNICAL FIELD

[0001] The present disclosure relates generally to wellbore operations and, more particularly (although not necessarily exclusively), to generating a dynamic multi-distribution model to facilitate a wellbore operation.

BACKGROUND

[0002] A wellbore can be formed in a subterranean formation for extracting produced hydrocarbon or other suitable material. A wellbore operation can be performed to form the wellbore and extract the produced hydrocarbon. The wellbore operation can include or otherwise involve generating inversion models to display wellbore characteristics, such as structural features or fluid boundaries, downhole in the wellbore or borehole.

[0003] The inversion models may be misinterpreted due to varying sensitivity of the inversion models to particular data. For example, a number of distribution curves used to generate an inversion model can affect which structural features or fluid boundaries are visible in the model. Additionally, the geology of the wellbore, frequency used by the tools, and other factors related to the data collection can affect the resulting appearance of an inversion model. The misinterpreted inversion models may cause mistakes in the wellbore operation, such as steering to a poorly chosen location, and may lead to inefficient or otherwise unsuccessful wellbore operations.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0004] FIG. 1 is a cross-sectional view of a well system with a downhole tool that can be used to collect data during a wellbore operation according to some examples of the present disclosure.

[0005] FIG. 2 is a block diagram of a computing system for generating a dynamic multi-distribution model to facilitate a wellbore operation according to some examples of the present disclosure.

[0006] FIG. 3 is an example of a user interface for displaying resistive properties of a geological formation using a dynamic multi-distribution model according to some examples of the present disclosure.

[0007] FIG. 4 is a flowchart of a process for generating a dynamic multi-distribution model to facilitate a wellbore operation according to some examples of the present disclosure.

DETAILED DESCRIPTION

[0008] Certain aspects and examples of the present disclosure relate to generating a dynamic multi-distribution model by combining at least two distribution outputs of a resistivity inversion model. The resistivity inversion model can be generated by performing inversion using downhole data (e.g., resistivity data) from a wellbore tool deployed in a wellbore. Inversion can be a mathematical process by which data is used to generate a model that is consistent with the data. The resistivity data can be measurements of electrical resistivity of a geological formation surrounding the wellbore. Thus, the resistivity inversion model can be a model which displays color-coded resistivity values for geological features (e.g., structural features or fluid boundaries) of the geological formation.

[0009] The distribution outputs of the resistivity inversion model may be produced by varying a number of distribution curves used to generate the resistivity inversion model. In this way, multiple, unique versions of the resistivity inversion model can be generated for the geological formation. That is, each distribution output of the inversion model may display different geological features or boundaries within the geological formation. For example, a distribution output

generated using a smaller number of distribution curves (e.g., 4 or 20) can be more sensitive to, and therefore display more dominantly, geological features with lower resistivity (e.g., clay or water-bearing formations). Conversely, a distribution output of the inversion model generated using a higher number of distribution curves (e.g., 100) can be more sensitive to, and therefore may display more dominantly, geological features with higher resistivity (e.g., hydrocarbon-bearing formations or dense rock).

[0010] Therefore, combining the distribution outputs of the resistivity inversion model to generate the dynamic multi-distribution output can provide a detailed and accurate depiction of the geological formation. Additionally, as a result of combining the distribution outputs, the dynamic multi-distribution output can show more geological features of the geological formation than any single inversion model. Thus, geological formations with varying resistive properties can be accurately and conveniently displayed with the dynamic multi-distribution model. The dynamic multi-distribution model can then be used to identify the geological features of the geological formation, adjust a wellbore operation (e.g., a drilling operation) performed at the wellbore, or a combination thereof.

[0011] In some examples, a computing device can generate the dynamic multi-distribution model. The dynamic multi-distribution model of an inversion can utilize existing resistivity data collected by a wellbore tool for an interval of a wellbore and multiple distribution outputs generated using the resistivity data. The distribution outputs can be versions of a resistivity inversion model generated with different numbers of distribution curves (e.g., anywhere between 1 and 100 distribution curves). The number distribution outputs created for a resistivity inversion model can be directly proportional to an accuracy of a corresponding dynamic multi-distribution model. That is, the more versions of a resistivity inversion model that are generated and combined to create a dynamic multi-distribution model, the more accurately the dynamic multi-distribution model can represent a wellbore's geological profile.

[0012] As noted above, the distribution outputs can each be sensitive to different data. Thus, each distribution output can provide a unique visual representation of a geological profile of the interval of the wellbore. Additionally, quality control (QC) factors can be evaluated for each distribution output. The QC factors can be evaluated by performing quality control operations, generating data related to quality control, or a combination thereof. For example, a misfit analysis can be performed on each distribution output to quantify a difference between resistivity values of each distribution output and the resistivity data collected at the interval of the wellbore. Then, based on geological profiles of each distribution output and on the QC factor evaluations, sections of each distribution output and can be selected and combined to generate the dynamic multi-distribution model.

[0013] By combining the sections of the distribution outputs, areas of minor and major resistivity contrast in the geological profile of the interval of the wellbore can be displayed in a single model. Moreover, noisy areas of the distribution outputs can be left out of when selecting sections of the distribution outputs to combine. Thus, the dynamic multi-distribution model can efficiently and accurately display a comprehensive visual representation of the geological profile of the interval of the wellbore. The dynamic multi-distribution model can then be used to efficiently interpret resistive properties of (e.g., identify) geological features associated with the interval of the wellbore, adjust a wellbore operation (e.g., a drilling operation) at the wellbore, or a combination thereof.

[0014] Illustrative examples are given to introduce the reader to the general subject matter discussed herein and are not intended to limit the scope of the disclosed concepts. The following sections describe various additional features and examples with reference to the drawings in which like numerals indicate like elements, and directional descriptions are used to describe the illustrative aspects, but, like the illustrative aspects, should not be used to limit the present disclosure.

[0015] FIG. 1 is a cross-sectional view of a well system **100** with a downhole tool **110** that can be used to collect data during a wellbore operation according to some examples of the present disclosure. The well system **100** can include a wellbore **102** extending through various earth strata. The wellbore **102** can be formed below a well surface **116** within a geological formation **104** that can include hydrocarbon material such as oil, gas, coal, or other suitable material. The geological formation **104** may be formed from geological material such as sand or shale. The geological formation **104** can also include layers of different geological material or pockets of different geological material. Additionally, the wellbore **102** or a well path of the wellbore **102** may be substantially vertical or horizontal with respect to the well surface **116**.

[0016] The wellbore **102** can include a casing string **108** or other suitable components, such as tubing string, a work string, etc., for accessing the wellbore **102**. The casing string **108** can be constructed from steel or other suitable material. The casing string **108** can be coupled to walls of the wellbore **102** via a sheath **106** made of material such as cement or other suitable coupling material. The sheath **106** can seal off disadvantageous formations, such as flowing salt or fractured formations, in the geological formation **104**. The downhole tool **110** can be positioned in the wellbore **102** via the casing string **108** or another suitable component that can deploy the downhole tool **110** in the wellbore **102**. In some examples, the casing string **108** can deploy the downhole tool **110** via a surface component **112**, such as a winch or other suitable component that can lower or otherwise deploy the downhole tool **110** into the wellbore **102**.

[0017] The downhole tool **110** can be used to collect downhole data. The downhole data can provide information regarding wellbore operation parameters (e.g., wellbore trajectory, drill bit orientation, drill bit revolutions per minute (RPM), weight on bit, etc.), the wellbore **102**, the geological formation **104**, or a combination thereof. In some examples, the downhole tool **110** may be a measurement-while-drilling (MWD) tool, a logging while drilling (LWD) tool, or a wireline logging tool. The MWD and LWD tools can be devices for incorporating measurement tools into the casing string **108** and can provide substantially contemporaneous (real-time) downhole data. The wireline logging tool can be a device for incorporating measurement tools into a wireline cable, which can be deployed in a drilled section of the wellbore **102**. In some examples, the downhole data can be resistivity data **220a-b** collected by the LWD tool or the wireline logging tool.

[0018] A computing device **114** can receive the downhole data from the downhole tool **110**. In some examples, the computing device **114** can be positioned downhole in the wellbore **102**, remote from the well system **100**, or in other suitable locations with respect to the well system **100**. The computing device **114** can be communicatively coupled to the downhole tool **110**, a wireline, other suitable components of the well system **100**, or a combination thereof, via a wired or wireless connection. The computing device **114** can be on or above a well surface **116** where the wellbore **102** begins.

[0019] Additionally, in some examples, the downhole data can be used to adjust a wellbore operation or subsequent wellbore operations. For example, the computing device **114** can generate a dynamic multi-distribution model using the downhole data. The dynamic multi-distribution model can be a visual representation of a portion of the geological formation **104** surrounding an interval **118** of the wellbore **102**. Thus, the dynamic multi-distribution model can be used to detect (e.g., identify) particular layers, rock formations, hydrocarbons, or other suitable geological features of the portion of the geological formation **104**. A human operator or the computing device **114** can then determine an adjustment to the wellbore operation (e.g., a change to a well path of the wellbore **102** during a drilling operation) based on the detected geological features. Additionally or alternatively, the human operator or the computing device **114** can determine one or more drilling parameters (e.g., a weight on bit, a rate of penetration, bit RPM, etc.) based on the detected geological features. The computing device **114** may further automatically control the wellbore operation (e.g., by implementing the change to the well path or the one or more drilling

parameters).

[0020] FIG. 2 is a block diagram of a computing system **200** for generating a dynamic multi-distribution model **214** to facilitate a wellbore operation according to some examples of the present disclosure. FIG. 2 is described with reference to components shown in FIG. 1.

[0021] The computing system **200** may include the computing device **114** that can use downhole data (e.g., resistivity data **220a-b**) to generate a dynamic multi-distribution model **214** of an interval of a wellbore **102**. The computing device **114** can include a processing device **202**, a power source **208**, an input/output device **206**, and a memory **216** communicatively coupled via a bus **204**. The components of the computing device **114** can be parts of a same computing device, or they can be distributed from one another. The input/output device **206** can include a display device **210**. Examples of the display device **210** can include a touchscreen display or a computer monitor. The display device **210** can be portable or fixed to a stationary object, such as a monitor arm. In some examples, the power source **208** may include a battery or an electrical cable.

[0022] The processing device **202** can include one processing device or multiple processing devices. Non-limiting examples of the processing device **202** include a Field-Programmable Gate Array (FPGA), an application-specific integrated circuit (ASIC), a micro processing device, etc. The processing device **202** can execute instructions **218** stored in the memory **216** to perform operations. In some examples, the instructions **218** can include processing device-specific instructions generated by a compiler or an interpreter from code written in any suitable computer-programming language, such as C, C++, C #, etc.

[0023] The processing device **202** can be communicatively coupled to the memory **216** via the bus **204**. The memory **216** can include one memory or multiple memories and can include a memory device. The memory **216** can be non-volatile and may include any type of memory that retains stored information when powered off. Non-limiting examples of the memory **216** include electrically erasable and programmable read-only memory (EEPROM), flash memory, or any other type of non-volatile memory. At least some of the memory can include a non-transitory computer-readable medium from which the processing device **202** can read instructions **218**. The non-transitory computer-readable medium can include electronic, optical, magnetic, or other storage devices capable of providing the processing device **202** with computer-readable instructions or other program code. Examples of the non-transitory computer-readable medium include magnetic disk(s), memory chip(s), ROM, random-access memory (RAM), an ASIC, a configured processing device, optical storage, or any other medium from which a computer processing device can read the instructions **218**.

[0024] In some examples, the memory **216** can include the instructions **218** for causing the processing device **202** to perform operations. For example, the processing device **202** can receive downhole data from a downhole tool **110** deployed in a wellbore **102** during a wellbore operation (e.g., a drilling operation, well logging, well completion, well stimulation, or well intervention). The downhole tool **110** may collect and transmit the downhole data to the processing device **202** in real-time (e.g., while the wellbore operation is being performed). In a particular example, the downhole tool **110** can be a logging while drilling (LWD) tool, which may comprise one or more transmitters for transmitting electrical currents (e.g., electromagnetic waves) through a geological formation **104** surrounding an interval **118** of the wellbore **102**. The LWD tool can further comprise one or more receivers for receiving the electromagnetic waves that have traveled through the geological formation and for measuring changes (e.g., attenuation or phase shift) in the electromagnetic waves from traveling through the geological formation **104**. The changes, such as attenuation and phase shift, can be directly related to the resistivity of the geological formation **104**. Thus, the LWD tool can determine resistivity data **220a** based on the measured changes in the electromagnetic waves. The resistivity data **220a** can therefore be indicative of the resistive properties of various layers, rock formations, fluids, or the like within the geological formation **104**.

[0025] The processing device **202** can further execute a resistivity inversion algorithm **224** to generate a resistivity inversion model **215** using the resistivity data **220a** collected by the downhole tool **110**. The resistivity inversion algorithm **224** can involve the processing device **202** generating a start model using synthetic resistivity data based on a theoretical data collection. The theoretical data collection can occur by simulating electrical current flow through expected geological features of the interval **118** of the wellbore **102**. The start model can then be adjusted by comparing the resistivity data **220a** with the synthetic resistivity data. The processing device **202** can repeatedly generate new synthetic resistivity data and can repeatedly adjust the start model by comparing the resistivity data **220a** with the new synthetic resistivity data. Comparing the resistivity data **220a** to the new synthetic resistivity data can improve the fit between the resistivity data **220a** and the synthetic resistivity data used for the resistivity inversion model **215**. Once a difference between the resistivity data **220a** and the synthetic resistivity data is minimized, the synthetic resistivity data can be used to create the resistivity inversion model **215**.

[0026] The resistivity inversion algorithm **224** can further involve generating distribution outputs **216a-d** of the resistivity inversion model **215**. For example, the resistivity inversion algorithm **224** can use different numbers of distribution curves to generate multiple versions of the resistivity inversion model **215**. In the particular example, a first distribution output **216a** of the resistivity inversion model **215** can be generated with ten distribution curves, a second distribution output **216b** can be generated with twenty distribution curves, a third distribution output **216c** can be generated with forty distribution curves, and a fourth distribution output **216d** can be generated with fifty distribution curves.

[0027] The number of distribution curves employed by the resistivity inversion algorithm **224** can affect which geological features are detected and depicted in each distribution output. For example, when utilizing a lower number of distribution curves, each curve in the resulting model can represent a broader range of geological features. Thus, the models characterized by a lower number of distribution curves (e.g., the first distribution output **216a**) may be more significantly influenced by larger, more continuous geological features. In many examples, the larger, more continuous features are lower in resistivity (e.g., clay or water-bearing formations). Conversely, when utilizing a larger number of distribution curves, each curve represents a narrower range of geological features. Thus, the resulting model (e.g., the fourth distribution output **216d**) may be more detailed (e.g., can show smaller or more isolated geological features). In many examples, the smaller or isolated geological features are more resistive (e.g., hydrocarbon-bearing formations or dense rock). Thus, each of the distribution outputs **216a-d** can show a different range of geological features, thereby each providing a unique visual representation of the geological formation **104**.

[0028] Additionally, for each of the distribution outputs **216a-d**, the processing device **202** can perform quality control (QC) operations. In doing so, the processing device **202** may generate QC data **228** for each distribution output **216a-d**. For example, the processing device **202** can perform a misfit analysis on the distribution outputs **216a-d**. The misfit analysis can involve determining a degree to which synthetic resistivity data matches corresponding, observed resistivity data. Thus, as a result of the misfit analysis, the QC data **228** can include one or more difference values representative of a difference between the resistivity data **220a** and the final synthetic data set used for each distribution output **216a-d**. The processing device **202** may further perform an error analysis to evaluate potential errors in data acquisition, processing, and inversion, which may arise from measurement errors, environmental noise, or processing assumptions. Thus, the QC data **228** can also include error or confidence values for the synthetic resistivity data, the resistivity data **220a**, or a combination thereof. In some examples, the QC data **228** can include also drilling logs or other suitable geophysical data collected and a QC operation can involve comparing the drilling logs or geophysical data to the synthetic resistivity data. In yet another example, the QC operations can involve performing missing data analysis by comparing the resistivity data **220a** or the synthetic resistivity data to data sets from different sources or times. The missing data analysis can

further involve generating a completeness ratio for the resistivity data **220a** or the synthetic resistivity data. The completeness ratio can be a measure of the amount of data available vs a total expected data.

[0029] After generating the distribution outputs **216a-d**, the processing device **202** can select a section **222a-d** from each distribution output **216a-d**. Each section **222a-d** selected can correspond to a different segment of the interval **118** of the wellbore **102**. In some examples, the processing device **202** may select the sections **222a-d** based on QC operations. For example, the processing device **202** can select sections that are unlikely to be missing a significant amount of data. In another example, the processing device **202** can select sections corresponding to low error values, low differences between synthetic and observed data, or corresponding to other desirable QC data **228**. In some examples, a machine learning model can be trained to select sections of distribution outputs based on the QC data **228**. In such examples, the machine learning model may be trained using various distribution outputs of various inversion models as well as QC data and section selections corresponding to each distribution output of each inversion model.

[0030] The processing device **202** can further generate a dynamic multi-distribution model **214** that includes the sections **222a-d** selected from each distribution output **216a-d**. For example, the processing device **202** can combine each section **222a-d** in an order corresponding to depth in the wellbore **102** to generate the dynamic multi-distribution model **214**. An example of combining the sections **222a-d** to generate the dynamic multi-distribution model **214** is described and depicted with respect to FIG. 3.

[0031] The processing device **202** can additionally output the dynamic multi-distribution model **214** via a user interface **212** at the display device **210**. The processing device **202** can then detect geological features surrounding the interval **118** of the wellbore **102** based on the dynamic multi-distribution model **214**. The processing device **202** can further determine an adjustment to the wellbore operation (e.g., an adjustment to a well path **232** of the wellbore **102**) or can determine one or more drilling parameters **230** based on the detected geological features. Additionally, the processing device **202** may control the wellbore operation (e.g., by implementing the drilling parameters **230** or by adjusting the well path **232**). The drilling parameters **230** may include a weight on bit, a rate of penetration, bit revolutions per minute, pump rate, rig power, etc.

[0032] Additionally, in some examples, the downhole tool **110** can be continuously collecting downhole data. Consequently, the processing device **202** may receive, from the downhole tool **110**, second resistivity data **220b** for the geological formation **104** associated with the interval **118** of the wellbore **102**. In response, the processing device **202** can execute the resistivity inversion algorithm **224** again to generate new distribution outputs for the resistivity inversion model **215**. When executing the resistivity inversion algorithm **224** again, the processing device **202** may input the first resistivity data **220a** and the second resistivity data **220b** to the resistivity inversion algorithm **224**. Thus, the new distribution outputs can be based on the first and second resistivity data **220a-b**. In other examples, the new distribution outputs may be based only on the second resistivity data **220b**.

[0033] The processing device **202** may then perform the QC operations on each of the new distribution outputs. Based on the QC operations, the processing device **202** may determine that, for example, a section of a new distribution output is more accurate than a first section **222a** of the first distribution output **216a**. In another example, based on the QC operations, the processing device **202** may determine that, the section of the new distribution output is missing less data than the first section **222a** of the first distribution output **216a**. In response, the processing device **202** can update a segment of the dynamic multi-distribution model **214** that comprises the first section **222a** from the first distribution output **216a**. In doing so, the processing device **202** can replace the segment of the dynamic multi-distribution model **214** with the section of the new distribution output.

[0034] The processing device **202** may then output, via the user interface **212**, the updated dynamic

multi-distribution model with the section of the new distribution output. Therefore, as new data becomes available due to the downhole tool **110** continuously collecting and transmitting data from the wellbore **102** in real-time, the dynamic multi-distribution model **214** can be updated. In this way, the dynamic multi-distribution model **214** can be dynamic such that it can provide an up-to-date and accurate visual representation of the geological formation **104** surrounding the wellbore **102**.

[0035] FIG. **3** is an example of a user interface **212** for displaying resistive properties of a geological formation **104** associated with a wellbore **102** using a dynamic multi-distribution model **214** according to some examples of the present disclosure. FIG. **3**. is described with references to the components shown in FIGS. **1-2**.

[0036] The dynamic multi-distribution model **214** can be output for display by the processing device **202** via the user interface **212** after the dynamic multi-distribution model **214** is generated. The dynamic multi-distribution model **214** shown in FIG. **3** includes a visual indicator **302** of a well path of the wellbore **102**. As depicted, the visual indicator **302** can be a line. In other examples, the visual indicator **302** can be rectangular, cylindrical, or any other suitable shape usable to illustrate the well path. The well path can be substantially horizontal or vertical with respect to a well surface, such as the well surface **116** depicted in FIG. **1**. The visual indicator **302** may or may not be in the same orientation (e.g., vertical or horizontal) as the well path.

[0037] Additionally, as depicted in FIG. **3**, the processing device **202** may output, via the user interface **212**, the distribution outputs **216a-d** of the resistivity inversion model **215**. In other examples, the user interface **212** may not include the distribution outputs **216a-d** and, instead, the processing device **202** may exclusively output the dynamic multi-distribution model **214**. Each distribution output **216a-d** can be characterized by a number of distribution curves. Thus, in some examples, the processing device **202** may output an indication of the number of distribution curves used for each distribution output **216a-d**. For example, as shown in FIG. **3**, the user interface **212** includes a first indicator **306a** that the first distribution output **216a** was generating using ten distribution curves, a second indicator **306b** that the second distribution output **216b** was generated using twenty distribution curves, a third indicator **306c** that the third distribution output **216c** was generated using forty distribution curves, and a fourth indicator **306d** that the fourth distribution output **216d** was generated using fifty distribution curves.

[0038] Additionally, the distribution outputs **216a-d** can include one or more regions delineated by different colors, patterns, other suitable color properties such as saturation, opacity, hue, brightness, or other suitable visual differences that represent differences in resistivity data along an interval **118** of the wellbore **102** represented by the resistivity inversion model **215**. Additionally, as discussed above with respect to FIG. **2**, sections **222a-d** of each distribution output **216a-d** can be selected and combined to generate the dynamic multi-distribution model **214**. Therefore, the dynamic multi-distribution model **214** can be made up of segments **310a-d** corresponding to the sections **222a-d** selected from each distribution output **216a-d**. In some examples, the user interface **212** can include an indication (e.g., arrows **312a-d**) of the which distribution output **216a-d** corresponds to each segment **310a-d** of the dynamic multi-distribution model **214**.

[0039] Moreover, as a result of combining the sections **222a-d** of the distribution outputs **216a-d** to create the dynamic multi-distribution model **214**, the dynamic multi-distribution model **214** can include unique visual representations from each of the distribution output sections. That is, the dynamic multi-distribution model **214** can include one or more regions delineated by different colors, patterns, other suitable color properties such as saturation, opacity, hue, brightness, or other suitable visual differences that represent the differences in the resistivity data for the interval **118** of the wellbore **102**.

[0040] For example, as depicted in FIG. **3**, relatively high resistivity data of the geological formation **104** in the dynamic multi-distribution model **214** can be represented by a first region **308a**. Additionally, relatively low resistivity data of the geological formation **104** in dynamic

multi-distribution model **214** can be represented by a second region **308b** and relatively moderate resistivity data of the geological formation **104** in the dynamic multi-distribution model **214** can be represented by a third region **308c**. Based on the visual differences between the first region **308a**, the second region **308b**, and the third region **308c**, the first region **308a** may be determined to have a highest resistivity and the second region **308b** may be determined to have the lowest.

[0041] The dynamic multi-distribution model **214** can further include depths for the interval **118** of the wellbore **102** on the x-axis. The dynamic multi-distribution model **214** can also include intervals **304a-b** on either side of the visual indicator **302** for the well path. The regions **306a-c** can be within the intervals **304a-b**. The intervals **304a-b** can represent a distance into the geological formation **104** on either side of the wellbore **102**. For example, the intervals **304a-b** can represent a distance perpendicular from the wellbore of up to one hundred feet. A wellbore operation or subsequent wellbore operations can then be directed according to the visual representations of resistivity for the geological formation **104** (e.g., regions **306a-c**) and additional information such as the depth or the distance perpendicular from the wellbore **102**.

[0042] FIG. **4** is a flowchart of a process **400** for generating a dynamic multi-distribution model **214** to facilitate a wellbore operation according to some examples of the present disclosure. While FIG. **4** depicts a certain sequence of steps for illustrative purposes, other examples can involve more steps, fewer steps, different steps, or a different order of the steps than is depicted in FIG. **4**. The process **400** is described with references to components shown in FIGS. **1-3**.

[0043] At block **402**, the process **400** can involve receiving, by a processing device **202** and from a downhole tool **110** deployed in a wellbore **102** during a drilling operation, resistivity data **220a** for a geological formation **104** associated with an interval **118** of the wellbore **102**. The downhole tool **110** can be a logging while drilling (LWD) tool. The LWD tool can automatically collect and transmit the resistivity data **220a** to the processing device **202** in real-time.

[0044] At block **404**, the process **400** can involve executing, by the processing device **202**, a resistivity inversion algorithm **224** to generate a plurality of distribution outputs **216a-d** of a resistivity inversion model **215** using the resistivity data **220a**. Each of the distribution outputs **216a-d** can comprise a unique visual representation of the geological formation **104** associated with the interval **118** of the wellbore **102**. For example, the resistivity inversion algorithm **224a** can use a different number of distribution curves to generate each distribution output **216a-d**. As a result, each distribution output **216a-d** can be a different version of the resistivity inversion model **215** that each show at least a slightly different range of geological features of the geological formation **104**.

[0045] At block **406**, the process **400** can involve selecting, by the processing device **202**, a section **222a-d** of each distribution output of the plurality of distribution outputs **216a-d**. Each of the sections **222a-d** selected from each of the distribution outputs **216a-d** can correspond with a different segment of the interval **118** of the wellbore **102**. For example, a first section **222a** can correspond to a first portion of the geological formation **104** surrounding from a first depth to a second depth within the interval **118** of the wellbore **102**. Then, a second section **222b** can correspond to a second portion of the geological formation **104** surrounding from the second depth to a third depth within the interval **118** of the wellbore **102**. The sections **222a-d** may be selected based on quality control (QC) operations performed by the processing device. For example, based on a misfit analysis, the processing device **202** can select the sections **222a-d** of each distribution output **216a-d** that best fit the resistivity data **220a**.

[0046] At block **408**, the process **400** can involve generating, by the processing device **202**, a dynamic multi-distribution model **214** comprising the sections **222a-d** selected from each distribution output of the plurality of distribution outputs **216a-d**. For example, the sections **222a-d** can be combined in an order corresponding to increasing depth with respect to the interval **118** of the wellbore **102**. The dynamic multi-distribution model **214** can therefore provide a comprehensive view of the interval **118** of the wellbore **102**.

[0047] At block **410**, the process **400** can involve outputting, by the processing device **202** and via a user interface **212**, the dynamic multi-distribution model **214**. The dynamic multi-distribution model **214** can be used to adjust the drilling operation. For example, the processing device **202** may determine and implement a change to a well path of the wellbore **102** based on the dynamic multi-distribution model **214**.

[0048] Illustrative examples are given to introduce the reader to the general subject matter discussed herein and are not intended to limit the scope of the disclosed concepts. The following sections describe various additional features and examples with reference to the drawings in which like numerals indicate like elements, and directional descriptions are used to describe the illustrative aspects, but, like the illustrative aspects, should not be used to limit the present disclosure.

[0049] In some aspects, a system, a method, and a non-transitory computer-readable medium for displaying resistivity properties of a subterranean formation using a dynamic distribution model are provided according to one or more of the following examples:

[0050] As used below, any reference to a series of examples is to be understood as a reference to each of those examples disjunctively (e.g., “Examples 1-4” is to be understood as “Examples 1, 2, 3, or 4”).

[0051] Example 1 is a system comprising: a processing device; and a non-transitory computer-readable medium that includes instructions executable by the processing device for causing the processing device to perform operations comprising: receiving, from a downhole tool deployed in a wellbore during a drilling operation, resistivity data for a geological formation associated with an interval of the wellbore; executing a resistivity inversion algorithm to generate a plurality of distribution outputs of a resistivity inversion model using the resistivity data, wherein each distribution output of the plurality of distribution outputs comprises a unique visual representation of the geological formation associated with the interval of the wellbore; selecting a section of each distribution output of the plurality of distribution outputs, wherein each section of each distribution output of the plurality of distribution outputs corresponds with a different segment of the interval of the wellbore; generating a dynamic multi-distribution model comprising the sections selected from each distribution output of the plurality of distribution outputs; and outputting, via a user interface, the dynamic multi-distribution model that is usable to adjust the drilling operation.

[0052] Example 2 is the system of example(s) 1, wherein the dynamic multi-distribution model comprises a first visual indicator representative of a well path of the interval of the wellbore, and wherein the dynamic multi-distribution model comprises the unique visual representation of the geological formation associated with the wellbore from each section of each distribution output of the plurality of distribution outputs.

[0053] Example 3 is the system of example(s) 1-2, wherein each distribution output of the plurality of distribution outputs is characterized by a number of distribution curves, wherein the dynamic multi-distribution model comprises a plurality of segments, and wherein the operation of outputting, via the user interface, the dynamic multi-distribution model further comprises: outputting an indication of the distribution output of the plurality of distribution outputs that corresponds to each segment of the plurality of segments of the dynamic multi-distribution model; and outputting the number of distribution curves for each distribution output of the plurality of distribution outputs from which the sections were selected for the dynamic multi-distribution model.

[0054] Example 4 is the system of example(s) 1-3, wherein the operations further comprise: performing, for each distribution output of the plurality of distribution outputs, a plurality of Quality Control (QC) operations.

[0055] Example 5 is the system of example(s) 1-4, wherein the operation of selecting the section of each distribution output of the plurality of distribution outputs is based at least in part on the plurality of QC operations.

[0056] Example 6 is the system of example(s) 1-5, wherein the resistivity data is first resistivity data, wherein the plurality of distribution outputs is a first plurality of distribution outputs, and wherein the operations further comprise: receiving, from the downhole tool deployed in the wellbore, second resistivity data for the geological formation associated with the interval of the wellbore; executing the resistivity inversion algorithm to generate a second plurality of distribution outputs for the resistivity inversion model using the first resistivity data and the second resistivity data; updating at least one section of the dynamic multi-distribution model based on at least one distribution output of the second plurality of distribution outputs; and outputting, via the user interface, the updated dynamic multi-distribution model.

[0057] Example 7 is the system of example(s) 1-6, wherein the operations further comprise adjusting the drilling operation by adjusting a well path of the wellbore or by adjusting one or more drilling parameters associated with the drilling operation.

[0058] Example 8 is a computer-implemented method comprising: receiving, from a downhole tool deployed in a wellbore during a drilling operation, resistivity data for a geological formation associated with an interval of the wellbore; executing a resistivity inversion algorithm to generate a plurality of distribution outputs of a resistivity inversion model using the resistivity data, wherein each distribution output of the plurality of distribution outputs comprises a unique visual representation of the geological formation associated with the interval of the wellbore; selecting a section of each distribution output of the plurality of distribution outputs, wherein each section of each distribution output of the plurality of distribution outputs corresponds with a different segment of the interval of the wellbore; generating a dynamic multi-distribution model comprising the sections selected from each distribution output of the plurality of distribution outputs; and outputting, via a user interface, the dynamic multi-distribution model that is usable to adjust the drilling operation.

[0059] Example 9 is the computer-implemented method of example(s) 8, wherein the dynamic multi-distribution model comprises a first visual indicator representative of a well path of the interval of the wellbore, and wherein the dynamic multi-distribution model comprises the unique visual representation of the geological formation associated with the wellbore from each section of each distribution output of the plurality of distribution outputs.

[0060] Example 10 is the computer-implemented method of example(s) 8-9, wherein each distribution output of the plurality of distribution outputs is characterized by a number of distribution curves, wherein the dynamic multi-distribution model comprises a plurality of segments, and wherein outputting, via the user interface, the dynamic multi-distribution model further comprises: outputting an indication of the distribution output of the plurality of distribution outputs that corresponds to each segment of the plurality of segments of the dynamic multi-distribution model; and outputting the number of distribution curves for each distribution output of the plurality of distribution outputs from which the sections were selected for the dynamic multi-distribution model.

[0061] Example 11 is the computer-implemented method of example(s) 8-10, wherein the method further comprises: performing, for each distribution output of the plurality of distribution outputs, a plurality of QC operations.

[0062] Example 12 is the computer-implemented method of example(s) 8-11, wherein selecting the section of each distribution output of the plurality of distribution outputs is based at least in part on the plurality of QC operations.

[0063] Example 13 is the computer-implemented method of example(s) 8-12, wherein the resistivity data is first resistivity data, wherein the plurality of distribution outputs is a first plurality of distribution outputs, and wherein the method further comprises: receiving, from the downhole tool deployed in the wellbore, second resistivity data for the geological formation associated with the interval of the wellbore; executing the resistivity inversion algorithm to generate a second plurality of distribution outputs for the resistivity inversion model using the first resistivity data and

the second resistivity data; updating at least one section of the dynamic multi-distribution model based on at least one distribution output of the second plurality of distribution outputs; and outputting, via the user interface, the updated dynamic multi-distribution model.

[0064] Example 14 is the computer-implemented method of example(s) 8-13, wherein the method further comprises adjusting the drilling operation by adjusting a well path of the wellbore or by adjusting one or more drilling parameters associated with the drilling operation.

[0065] Example 15 is a non-transitory computer-readable medium comprising instructions that are executable by a processing device for causing the processing device to perform operations comprising: receiving, from a downhole tool deployed in a wellbore during a drilling operation, resistivity data for a geological formation associated with an interval of the wellbore; executing a resistivity inversion algorithm to generate a plurality of distribution outputs of a resistivity inversion model using the resistivity data, wherein each distribution output of the plurality of distribution outputs comprises a unique visual representation of the geological formation associated with the interval of the wellbore; selecting a section of each distribution output of the plurality of distribution outputs, wherein each section of each distribution output of the plurality of distribution outputs corresponds with a different segment of the interval of the wellbore; generating a dynamic multi-distribution model comprising the sections selected from each distribution output of the plurality of distribution outputs; and outputting, via a user interface, the dynamic multi-distribution model that is usable to adjust the drilling operation.

[0066] Example 16 is the non-transitory computer-readable medium of example(s) 15, wherein the dynamic multi-distribution model comprises a first visual indicator representative of a well path of the interval of the wellbore, and wherein the dynamic multi-distribution model comprises the unique visual representation of the geological formation associated with the wellbore from each section of each distribution output of the plurality of distribution outputs.

[0067] Example 17 is the non-transitory computer-readable medium of example(s) 15-16, wherein each distribution output of the plurality of distribution outputs is characterized by a number of distribution curves, wherein the dynamic multi-distribution model comprises a plurality of segments, and wherein the operation of outputting, via the user interface, the dynamic multi-distribution model further comprises: outputting an indication of the distribution output of the plurality of distribution outputs that corresponds to each segment of the plurality of segments of the dynamic multi-distribution model; and outputting the number of distribution curves for each distribution output of the plurality of distribution outputs from which the sections were selected for the dynamic multi-distribution model.

[0068] Example 18 is the non-transitory computer-readable medium of example(s) 15-17, wherein the operations further comprise: performing, for each distribution output of the plurality of distribution outputs, a plurality of QC operations.

[0069] Example 19 is the non-transitory computer-readable medium of example(s) 15-18, wherein the operation of selecting the section of each distribution output of the plurality of distribution outputs is based at least in part on the plurality of QC operations.

[0070] Example 20 is the non-transitory computer-readable medium of example(s) 15-19, wherein the resistivity data is first resistivity data, wherein the plurality of distribution outputs is a first plurality of distribution outputs, and wherein the operations further comprise: receiving, from the downhole tool deployed in the wellbore, second resistivity data for the geological formation associated with the interval of the wellbore; executing the resistivity inversion algorithm to generate a second plurality of distribution outputs for the resistivity inversion model using the first resistivity data and the second resistivity data; updating at least one section of the dynamic multi-distribution model based on at least one distribution output of the second plurality of distribution outputs; and outputting, via the user interface, the updated dynamic multi-distribution model.

[0071] The foregoing description of certain examples, including illustrated examples, has been presented only for the purpose of illustration and description and is not intended to be exhaustive or

to limit the disclosure to the precise forms disclosed. Numerous modifications, adaptations, and uses thereof will be apparent to those skilled in the art without departing from the scope of the disclosure.

Claims

1. A system comprising: a processing device; and a non-transitory computer-readable medium that includes instructions executable by the processing device for causing the processing device to perform operations comprising: receiving, from a downhole tool deployed in a wellbore during a drilling operation, resistivity data for a geological formation associated with an interval of the wellbore; executing a resistivity inversion algorithm to generate a plurality of distribution outputs of a resistivity inversion model using the resistivity data, wherein each distribution output of the plurality of distribution outputs comprises a unique visual representation of the geological formation associated with the interval of the wellbore; selecting a section of each distribution output of the plurality of distribution outputs, wherein each section of each distribution output of the plurality of distribution outputs corresponds with a different segment of the interval of the wellbore; generating a dynamic multi-distribution model comprising the sections selected from each distribution output of the plurality of distribution outputs; and outputting, via a user interface, the dynamic multi-distribution model that is usable to adjust the drilling operation.
2. The system of claim 1, wherein the dynamic multi-distribution model comprises a first visual indicator representative of a well path of the interval of the wellbore, and wherein the dynamic multi-distribution model comprises the unique visual representation of the geological formation associated with the wellbore from each section of each distribution output of the plurality of distribution outputs.
3. The system of claim 1, wherein each distribution output of the plurality of distribution outputs is characterized by a number of distribution curves, wherein the dynamic multi-distribution model comprises a plurality of segments, and wherein the operation of outputting, via the user interface, the dynamic multi-distribution model further comprises: outputting an indication of the distribution output of the plurality of distribution outputs that corresponds to each segment of the plurality of segments of the dynamic multi-distribution model; and outputting the number of distribution curves for each distribution output of the plurality of distribution outputs from which the sections were selected for the dynamic multi-distribution model.
4. The system of claim 1, wherein the operations further comprise: performing, for each distribution output of the plurality of distribution outputs, a plurality of Quality Control (QC) operations.
5. The system of claim 4, wherein the operation of selecting the section of each distribution output of the plurality of distribution outputs is based at least in part on the plurality of QC operations.
6. The system of claim 1, wherein the resistivity data is first resistivity data, wherein the plurality of distribution outputs is a first plurality of distribution outputs, and wherein the operations further comprise: receiving, from the downhole tool deployed in the wellbore, second resistivity data for the geological formation associated with the interval of the wellbore; executing the resistivity inversion algorithm to generate a second plurality of distribution outputs for the resistivity inversion model using the first resistivity data and the second resistivity data; updating at least one section of the dynamic multi-distribution model based on at least one distribution output of the second plurality of distribution outputs; and outputting, via the user interface, the updated dynamic multi-distribution model.
7. The system of claim 1, wherein the operations further comprise adjusting the drilling operation by adjusting a well path of the wellbore or by adjusting one or more drilling parameters associated with the drilling operation.
8. A computer-implemented method comprising: receiving, from a downhole tool deployed in a

wellbore during a drilling operation, resistivity data for a geological formation associated with an interval of the wellbore; executing a resistivity inversion algorithm to generate a plurality of distribution outputs of a resistivity inversion model using the resistivity data, wherein each distribution output of the plurality of distribution outputs comprises a unique visual representation of the geological formation associated with the interval of the wellbore; selecting a section of each distribution output of the plurality of distribution outputs, wherein each section of each distribution output of the plurality of distribution outputs corresponds with a different segment of the interval of the wellbore; generating a dynamic multi-distribution model comprising the sections selected from each distribution output of the plurality of distribution outputs; and outputting, via a user interface, the dynamic multi-distribution model that is usable to adjust the drilling operation.

9. The computer-implemented method of claim 8, wherein the dynamic multi-distribution model comprises a first visual indicator representative of a well path of the interval of the wellbore, and wherein the dynamic multi-distribution model comprises the unique visual representation of the geological formation associated with the wellbore from each section of each distribution output of the plurality of distribution outputs.

10. The computer-implemented method of claim 8, wherein each distribution output of the plurality of distribution outputs is characterized by a number of distribution curves, wherein the dynamic multi-distribution model comprises a plurality of segments, and wherein outputting, via the user interface, the dynamic multi-distribution model further comprises: outputting an indication of the distribution output of the plurality of distribution outputs that corresponds to each segment of the plurality of segments of the dynamic multi-distribution model; and outputting the number of distribution curves for each distribution output of the plurality of distribution outputs from which the sections were selected for the dynamic multi-distribution model.

11. The computer-implemented method of claim 8, wherein the method further comprises: performing, for each distribution output of the plurality of distribution outputs, a plurality of QC operations.

12. The computer-implemented method of claim 11, wherein selecting the section of each distribution output of the plurality of distribution outputs is based at least in part on the plurality of QC operations.

13. The computer-implemented method of claim 8, wherein the resistivity data is first resistivity data, wherein the plurality of distribution outputs is a first plurality of distribution outputs, and wherein the method further comprises: receiving, from the downhole tool deployed in the wellbore, second resistivity data for the geological formation associated with the interval of the wellbore; executing the resistivity inversion algorithm to generate a second plurality of distribution outputs for the resistivity inversion model using the first resistivity data and the second resistivity data; updating at least one section of the dynamic multi-distribution model based on at least one distribution output of the second plurality of distribution outputs; and outputting, via the user interface, the updated dynamic multi-distribution model.

14. The computer-implemented method of claim 8, wherein the method further comprises adjusting the drilling operation by adjusting a well path of the wellbore or by adjusting one or more drilling parameters associated with the drilling operation.

15. A non-transitory computer-readable medium comprising instructions that are executable by a processing device for causing the processing device to perform operations comprising: receiving, from a downhole tool deployed in a wellbore during a drilling operation, resistivity data for a geological formation associated with an interval of the wellbore; executing a resistivity inversion algorithm to generate a plurality of distribution outputs of a resistivity inversion model using the resistivity data, wherein each distribution output of the plurality of distribution outputs comprises a unique visual representation of the geological formation associated with the interval of the wellbore; selecting a section of each distribution output of the plurality of distribution outputs, wherein each section of each distribution output of the plurality of distribution outputs corresponds

with a different segment of the interval of the wellbore; generating a dynamic multi-distribution model comprising the sections selected from each distribution output of the plurality of distribution outputs; and outputting, via a user interface, the dynamic multi-distribution model that is usable to adjust the drilling operation.

16. The non-transitory computer-readable medium of claim 15, wherein the dynamic multi-distribution model comprises a first visual indicator representative of a well path of the interval of the wellbore, and wherein the dynamic multi-distribution model comprises the unique visual representation of the geological formation associated with the wellbore from each section of each distribution output of the plurality of distribution outputs.

17. The non-transitory computer-readable medium of claim 15, wherein each distribution output of the plurality of distribution outputs is characterized by a number of distribution curves, wherein the dynamic multi-distribution model comprises a plurality of segments, and wherein the operation of outputting, via the user interface, the dynamic multi-distribution model further comprises: outputting an indication of the distribution output of the plurality of distribution outputs that corresponds to each segment of the plurality of segments of the dynamic multi-distribution model; and outputting the number of distribution curves for each distribution output of the plurality of distribution outputs from which the sections were selected for the dynamic multi-distribution model.

18. The non-transitory computer-readable medium of claim 15, wherein the operations further comprise: performing, for each distribution output of the plurality of distribution outputs, a plurality of QC operations.

19. The non-transitory computer-readable medium of claim 18, wherein the operation of selecting the section of each distribution output of the plurality of distribution outputs is based at least in part on the plurality of QC operations.

20. The non-transitory computer-readable medium of claim 15, wherein the resistivity data is first resistivity data, wherein the plurality of distribution outputs is a first plurality of distribution outputs, and wherein the operations further comprise: receiving, from the downhole tool deployed in the wellbore, second resistivity data for the geological formation associated with the interval of the wellbore; executing the resistivity inversion algorithm to generate a second plurality of distribution outputs for the resistivity inversion model using the first resistivity data and the second resistivity data; updating at least one section of the dynamic multi-distribution model based on at least one distribution output of the second plurality of distribution outputs; and outputting, via the user interface, the updated dynamic multi-distribution model.
